ALEXANDER TONG

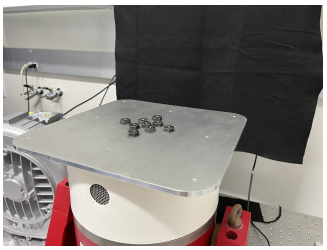
Mechanical engineering | Recovery hardware and supersonic aerodynamics

Boston University, B.S. Mechanical Engineering, Aerospace Concentration | Expected May 2028
alto335@bu.edu | 860-706-3624 | alexandertong.space

PORTFOLIO AT A GLANCE

Project Index

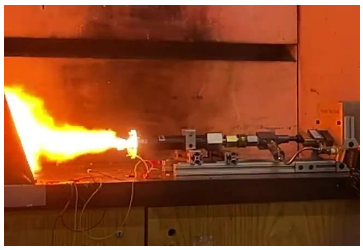
Page 2 / Separation & recovery	Earlier black-powder configuration: 18.7 lbm within a 20 lbm allocation. Load sizing, CO2 redesign and an operational vacuum chamber.
Page 3 / Aerodynamic fairing	Three concept studies; Ansys CFD baseline converged and under review. Shared aerodynamic analysis, integration and manufacturing trade.
Pages 4-5 / Plummer shaker hardware	Structural analysis, bored-bolt checks and CNC fabrication. Installed plate-and-bolt hardware at 93.9% lower estimated cost.



Plummer: machined adapter installed.



AIAA: assembled Prototype 1 aircraft.



Hybrid nozzle: hotfire test frame.

Additional Projects

Plummer Laboratory

CNC-machined shaker adapter; plate-and-bolt hardware estimate reduced from $1,170 to $71.11, excluding labor. Designed 48 Instron tests; verified coupling-coefficient sign and behavior, without a numerical magnitude.

AIAA Design-Build-Fly wing

Complete Prototype 1 wing 42.7% lighter than the previous complete wing; build time reduced from three months to one. Led design, fabrication and member training.

Hybrid rocket nozzle

73% mass reduction from my initial design; approximately two-second hotfire. Minimum yield FOS 7.8 in SolidWorks FEA for mechanical loads only; thermal effects excluded. Thrust gain unmeasured.

EK210 line-following car

Designed and programmed the vehicle: six loaded course runs in 70-86 seconds against a 90-second limit. Recorded component cost $73.94 against a $200 budget.

Culturon

Executed the company's first full-scale nanoparticle deposition map using microscopy and ImageJ. Designed adhesive-free sample racks; presented prevention concepts. Public details limited by NDA.

Manufacturing & mentoring

CNC and manual machining, hot-wire foam cutting and composite fabrication. Taught BURPG members turning and milling through the machinist's-hammer project beginning Fall 2025.

Additional research & coursework

Space Physics & Technology Lab: instrument-data analysis with Python/pySPEDAS and assistance with flight-spare laser calibration. University of Sydney pulsed plasma thruster: inductor-spool design and coil-winding calculations. ME360 course project includes completed linear-stage/CNC drawing and fishing-game builds.

01 / RECOVERY HARDWARE

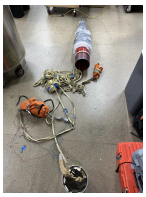
Argo Separation and Recovery Subsystem

Boston University Rocket Propulsion Group | Subsystem Co-Lead | June 2025-present

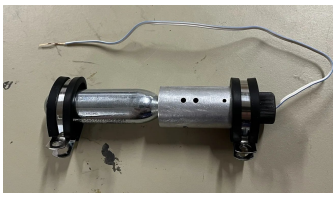
18.7 / 20 lbm
Earlier BP mass / allocation; BP BOM $1,076

823 / 1,230 lbf
Opening-load / conservative snatch-load estimates

352 L
Operational vacuum chamber; reported test pressure 0.011 atm



Subsystem before CO2 coldflow; historical pretest photo.



Manufactured CO2 cartridge/piercer assembly. Design and fabrication shared by our three-person team.



Complete chamber design and door/adapter/seal analysis by me; fabrication support and DAQ development by my co-lead.

Design and Test Leadership

Developed Argo recovery hardware through four design reviews. Compared five load methods; selected 823 lbf opening and conservative 1,230 lbf snatch-load design estimates, informed by a limited historical Icarus comparison. Paired flight computers provide separate firing paths.

Authored the integrated vinyl-charge black-powder SOP and led the second integrated test; collaborated on inspection and failure assessment. One of three charges failed in each test; the second damaged the parachute bag and canopy.

Standalone successes did not establish integrated reliability. End-seal variability is the leading hypothesis; adjacent-charge damage was inferred. Root cause remains unconfirmed.

Designed and sized CO2 piercers using pressure and Lamé thick-wall analysis: calculated hoop-stress FOS 2.01 and thread-shear FOS 6.31. Contributed CAD, drawings, machining and component testing in a three-person team.

Calculation basis

Recovery lines	Selected 1/2 in, 6,000 lbf Kevlar stock. Published comparisons motivated splice selection: near-full strength retention, approximately 2× a 50%-efficient knot. Our strength gain remains unmeasured.
CO2 deployment sizing	Calculated 4.66 psi separation requirement for four nylon shear pins plus estimated interface friction. Ideal-gas estimate for the modeled free volume: 10.9 g versus a selected 16 g cartridge.
CO2 piercer housing	Calculated hoop-stress FOS 2.01; thread-shear FOS 6.31. Separate checks using 40 ksi typical yield; combined-stress nominal yield ratio ~1.37. Pressure basis and local stresses remain unverified.
Vacuum door / Roark	12 in unsupported diameter; 0.25 in 6061-T6 plate; 14.7 psi differential gives 1,663 lbf. Recorded bending stress 16.5 ksi; Roark deflection 0.0848 in; yield FOS 2.12 at 240 MPa yield strength. Uniform pressure, simply supported circumference. Stress is a separate calculator entry.
Legacy Catclaw body	Weighed mass reduction: 49.5%; piston-retention slot revised. Later Lamé ratio reported as 2.01; source and yield criterion under review. Separate from the CO2 piercer.

Verification Status

Component firing completed at reported 0.011 atm. Integrated trial: cartridge discharged; nosecone did not separate. A wiring-port leak prompted an NPT/epoxy/putty seal revision; retest and flight tests are pending. Splice trial: no observed slip at an estimated 1,800 lbf interval-average load; peak force and ultimate strength unmeasured. My co-lead developed DAQ and led pressure-data acquisition/analysis.

Recovery architecture, hardware & tests | CO2 sizing & stress calculations | Chamber design

02 / AERODYNAMICS

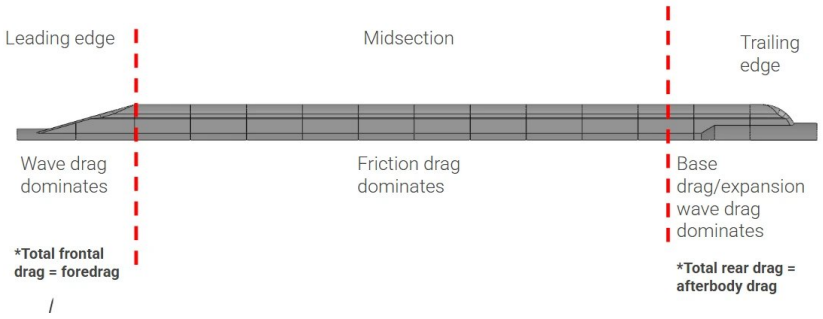
Argo Downcomer Aerocover

Boston University Rocket Propulsion Group | Project Lead | August 2026-present

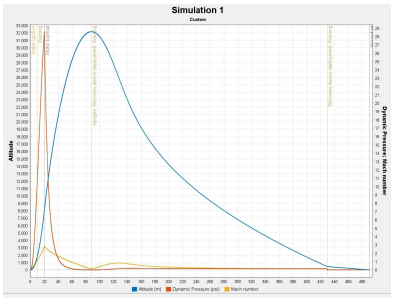
63 in × 1.6 in
Fairing length / nominal height on an 8 in diameter rocket

Mach 2.0-2.81
Priority comparison range from the simulated trajectory

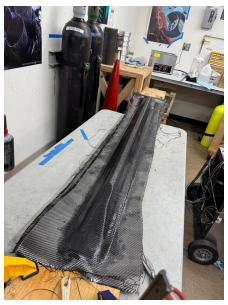
3 concepts
Teammate-owned geometries; shared requirements and analysis by me



My CoDR diagram: a first-principles decomposition, not a measured allocation of drag. Pressure and friction act across the whole fairing.



Trajectory used to select high-dynamic-pressure analysis conditions; simulation, not flight data.



Legacy Icarus fairing: the CFD baseline and a backup flight candidate, not a new design by me.

Analysis

Used D = q Sref CD to prioritize conditions by dynamic pressure as well as Mach. Evaluated leading-edge pressure drag, midsection wetted area and aft pressure recovery together with thermal loads, attachment integrity and manufacturing constraints. Isolated axisymmetric profile rankings cannot establish the performance of a partial fairing attached to the rocket body: junction flow and fin interference require integrated analysis.

Preliminary friction estimate	At Mach 2.81 and recorded Re = 2.04 × 10^8: $C_{D,f} \approx 0.0792$, using 477 in² wetted area, 7.99 in² reference area and a 10% preliminary roughness allowance.
Leading-edge surrogate	A 15° full-cone correlation gives $C_p \approx 0.1888$. It is a screening surrogate for the wedge-like legacy front, not the mounted fairing's total drag coefficient.

Contributions and Status

I developed the common analysis, integration/manufacturing trade and CoDR. Quinn Colleran, Hannah Ruan and Aaron Lin own the three concept geometries. My Mach 2.81 Icarus solve in **Ansys Fluent is converged and under review**; candidate ranking and Ansys adjoint optimization remain pending.

The investigation also led to a nine-profile-family nosecone sweep in RASAero II. A ranking difference from published comparisons prompted further investigation; it does not establish a better flight design.

Concept credits, hand calculations, sources & nosecone sweep | Hand-calculation reference: S. S. Chin, Missile Configuration Design (1961).

03 / BU PLUMMER LABORATORY

Shaker Adapter and Mounting Bolts

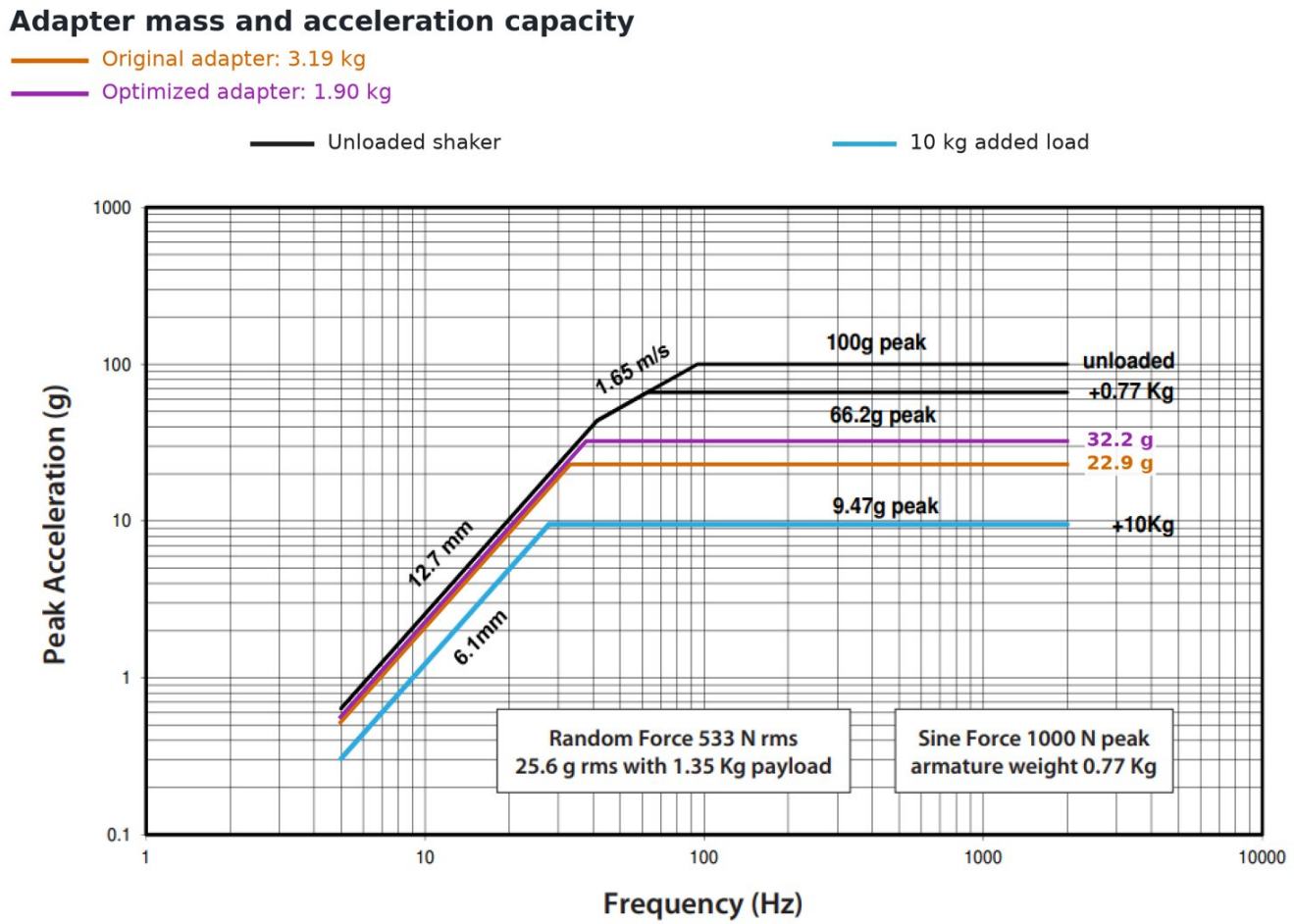
Mechanical Engineering Intern | June 2026-present

0.44 / 1 mm
Dynamic relative deflection / limit; conservative 3 kg payload

+40.7%
Calculated force-limited acceleration capacity; 0.50 kg payload

2.46 FOS
Plate bearing under the modeled combined bolt load

The adapter mounts a physical Ising-model fixture for controlled shaker vibration tests of bistable buckling elements.



Calculated adapter envelopes over the manufacturer's shaker chart; both include the same 0.50 kg scientific payload. The +40.7% gain applies to the force-limited plateau; lower-frequency capacity is constrained by stroke and velocity.

Design and Manufacturing

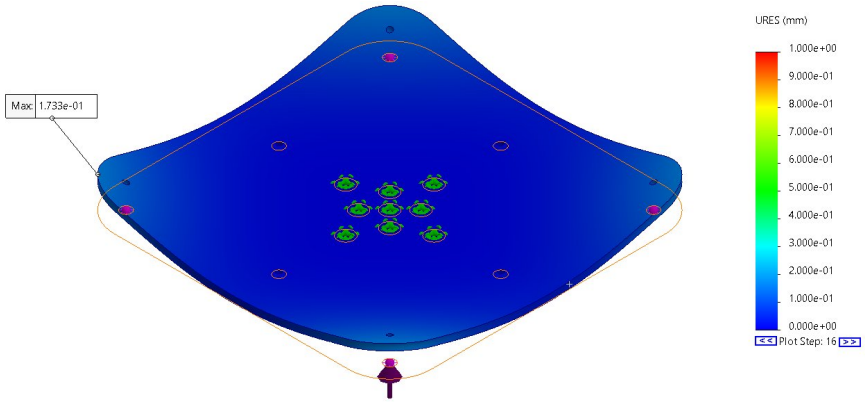
Replaced an estimated $1,170 outsourced plate/custom-bolt solution with $71.11 of in-house hardware. Developed SolidWorks CAD/FEA, drawings and CAM/G-code; machined the 6061-T6 plate on a Haas VF-2 and checked plate dimensions with calipers against the ±0.005 in drawing tolerance.

Optimized adapter geometry through hand calculations and SolidWorks FEA: 3.19 to 1.90 kg, a 40.4% mass reduction. With the same 0.50 kg scientific payload, calculated force-limited acceleration rises from 22.9 to 32.2 g (+40.7%), including the 0.77 kg built-in armature.

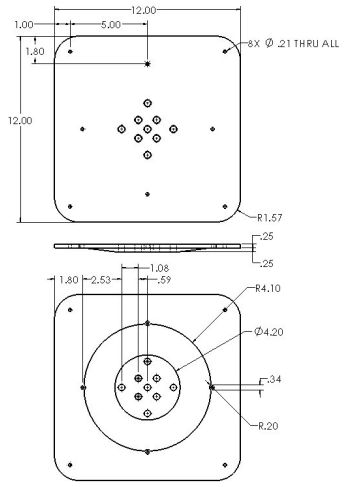
Drilled 5 mm axial bores in standard medium-carbon-steel M10 bolts for the required shaker interface after unsuccessful stainless-steel machining attempts. The bored bolts and adapter are installed.

Acceleration model and full-size graph | Structural calculations and FEA evidence continue on page 5.

Structural Verification and Manufacturing Evidence



SolidWorks plate-displacement result: 0.173 mm maximum. This is a model prediction.



My 12 × 12 in adapter drawing, including underside geometry.

Structural calculation basis

Installed joint geometry	M10 × 1.5 with a 5 mm axial bore; modeled remaining tensile area 38.35 mm². Nominal Class 8.8 yield strength 640 MPa; local plate bearing thickness 9.7 mm; thread engagement 9.35 mm.
Per-bolt load path	5 kg allowance (2 kg plate + 3 kg fixture) at 11 G, shared by four bolts: 134.9 N external load per bolt. Workbook conservatively adds the full external load to 16.36 kN preload: 16.50 kN combined. Torque estimate: 21.7 ft-lbf at K = 0.18; joint stiffness sharing is not modeled.
Static FOS by checked mode	Bolt tensile yield: 1.49 under preload plus full external load; preload alone gives 1.50. Plate bearing 2.46; pull-through 4.99; external-thread shear 3.71; internal-thread shear 3.60. These are mode-specific ratios, not an overall dynamic FOS.
Displacement checks	Review plate: 0.173 mm versus 1.27 mm L/240 limit. Dynamic fixture: 0.44 mm relative deflection versus 1 mm limit, with 3 kg on top at 20.1 Hz and 13 mm peak-to-peak base motion. These FEA cases are separate from the 0.50 kg acceleration comparison.

The bolt preload alone gives a 1.50 tensile-yield ratio; conservatively adding the full external load gives 1.49. While the joint stays clamped, only a fraction of added separating load raises bolt tension; the rest reduces clamping force. Joint stiffness and load introduction determine that fraction. The workbook does not quantify it, preload scatter, separation or fatigue. Displacements are predictions; cost covers plate and bolts only.

Full calculations, drawings, FEA and machining evidence | Sources: my design review and DSRF Calcs; joint interpretation: NASA TM-106943.